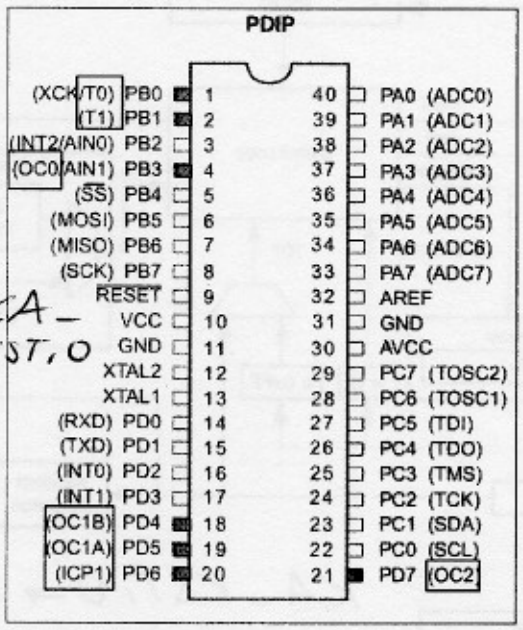
STRUKTURY WBUDOWANE

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Opracował: K. Arnold
KST, O
Politechnika Poznańska

Pinout ATmega16A



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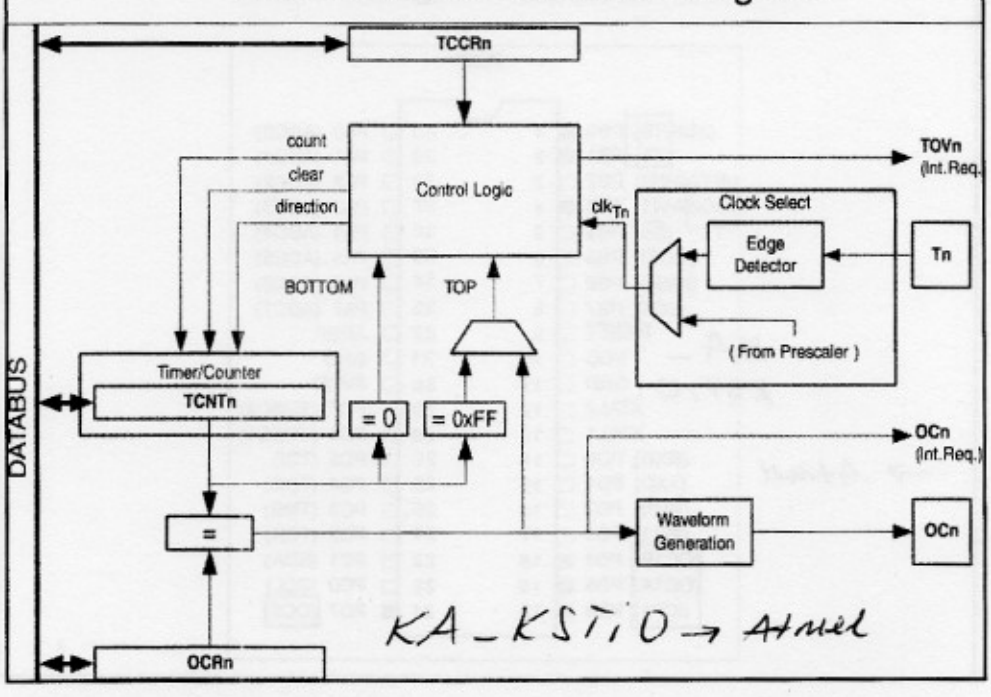
8-bit Timer/Counter0 with PWM

- Single Compare Unit Counter
- Clear Timer on Compare Match (Auto Reload)
- Glitch-free, Phase Correct Pulse Width Modulator (PWM)
- Frequency Generator
- External Event Counter
- 10-bit Clock Prescaler
- Overflow and Compare Match Interrupt Sources (TOV0 and OCF0)

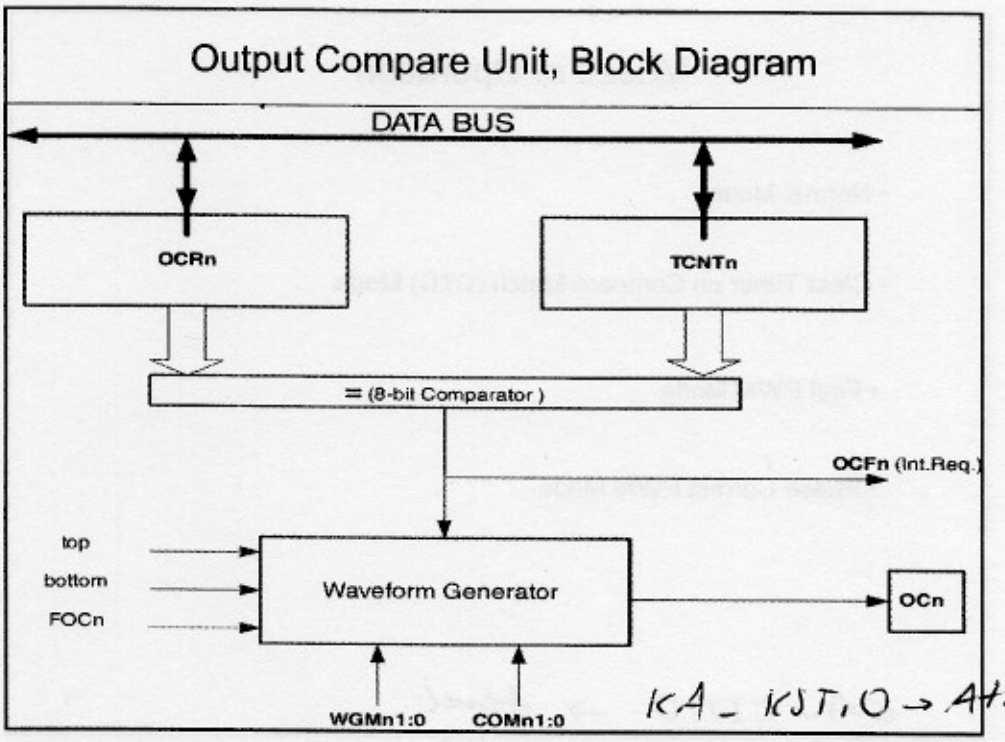
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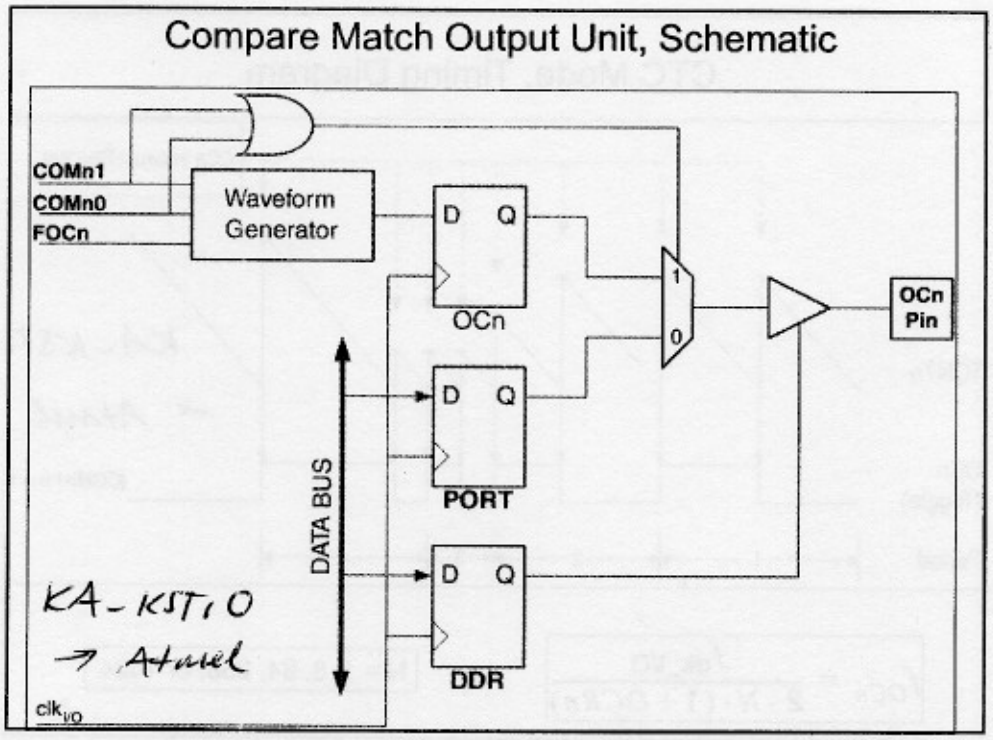
8-bit Timer/Counter Block Diagram



Output Compare Unit, Block Diagram



Compare Match Output Unit, Schematic



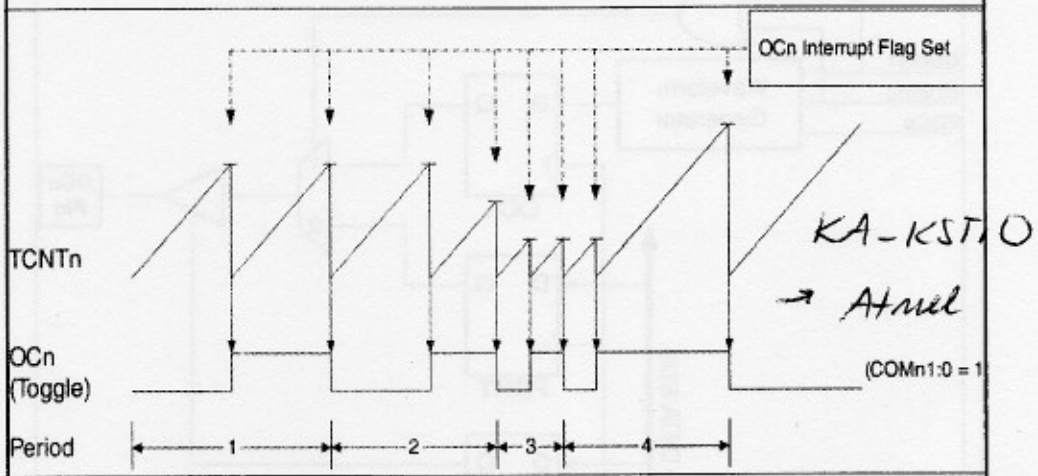
Modes of Operation

- Normal Mode
- Clear Timer on Compare Match (CTC) Mode
- Fast PWM Mode
- Phase Correct PWM Mode

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CTC Mode, Timing Diagram



$$f_{OCn} = \frac{f_{dk_I/O}}{2 \cdot N \cdot (1 + OCRn)}$$

N = 1, 8, 64, 256, or 1024

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CTC Mode, Output Frequency

$$f_{OCn} = \frac{f_{clk_I/O}}{2 \cdot N \cdot (1 + OCRn)}$$

N = 1, 8, 64, 256, or 1024

toggle

A maximum frequency

$f_{OC0} = f_{clk_I/O}/2$ when OCR0 is set to zero (0x00)

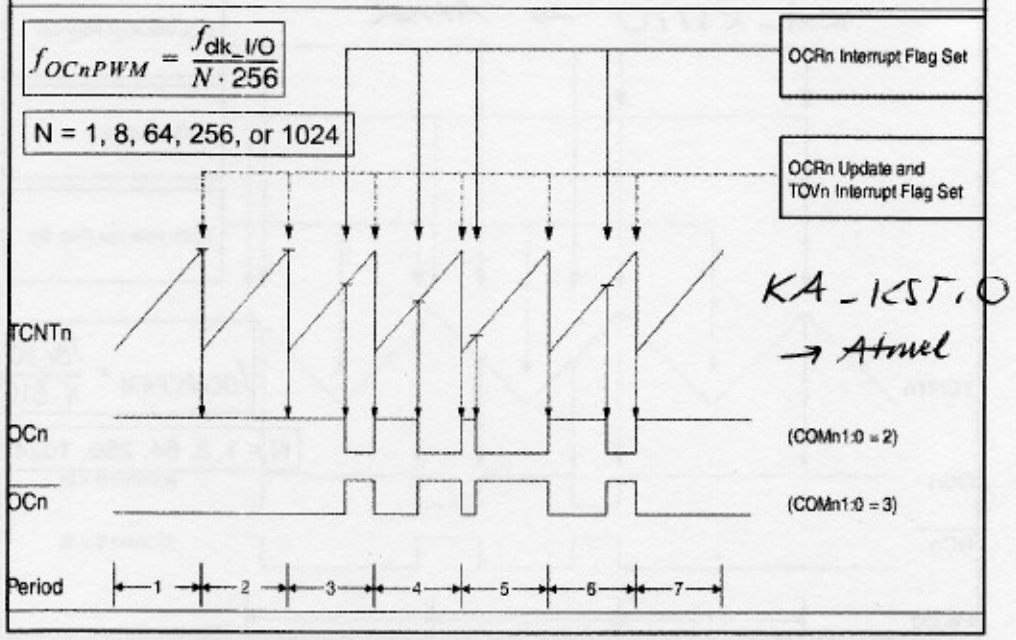
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Fast PWM Mode, Timing Diagram

$$f_{OCnPWM} = \frac{f_{clk_I/O}}{N \cdot 256}$$

N = 1, 8, 64, 256, or 1024



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Fast PWM Mode, Output Mode

Non-Inverting Compare Output Mode (COM01:0 = 2)

TCNT0 = OCR0 → OC0 Output clear
TCNT0 = BOTTOM → OCO Output set

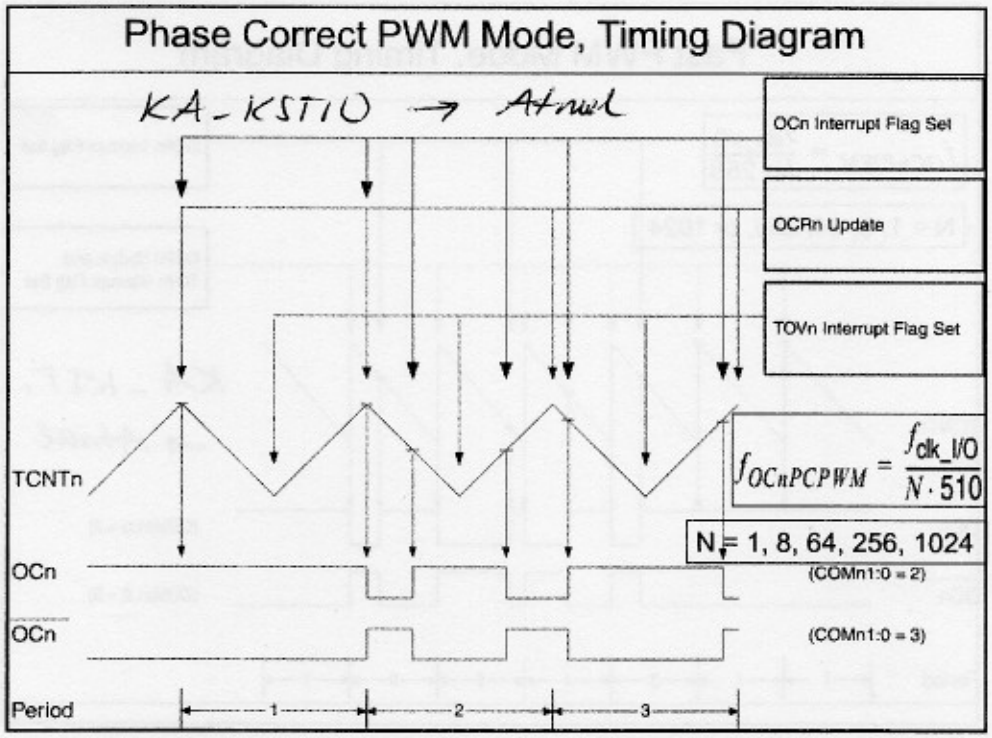
Inverting Compare Output Mode (COM01:0 = 3)

TCNT0 = OCR0 → OC0 Output set
TCNT0 = BOTTOM → OCO Output clear

KA-KST10

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Phase Correct PWM Mode, Timing Diagram



Phase Correct PWM Mode, Output Frequency

$$f_{OCnPCPWM} = \frac{f_{clk_I/O}}{N \cdot 510}$$

N = 1, 8, 64, 256, or 1024

The extreme values for the OCR0 Register

OCR0 = BOTTOM
- the output will be continuously low

OCR0 = MAX
- the output will be continuously high for non-inverted PWM mode. For inverted PWM the output will have the opposite logic values.

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Phase Correct PWM Mode, Output Mode

Non-Inverting Compare Output Mode (COM01:0 = 2)

TCNT0 = OCR0, counter increments → OC0 Output clear
TCNT0 = OCR0, counter decrements → OCO Output set

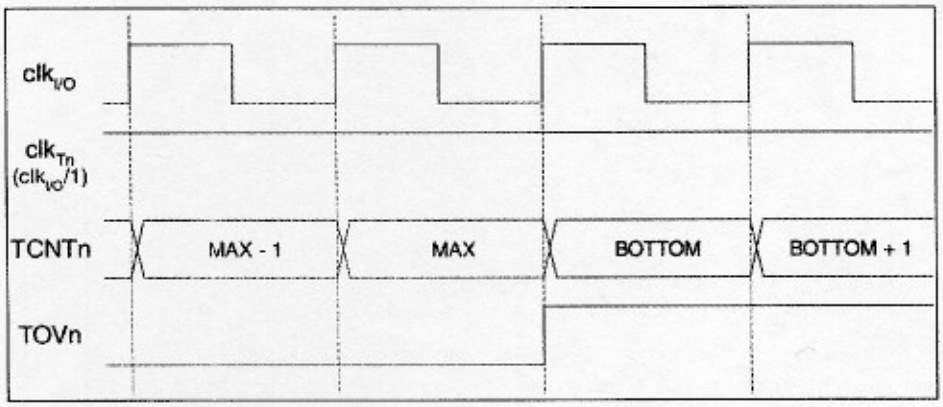
Inverting Compare Output Mode (COM01:0 = 3)

TCNT0 = OCR0, counter increments → OC0 Output set
TCNT0 = OCR0, counter decrements → OCO Output clear

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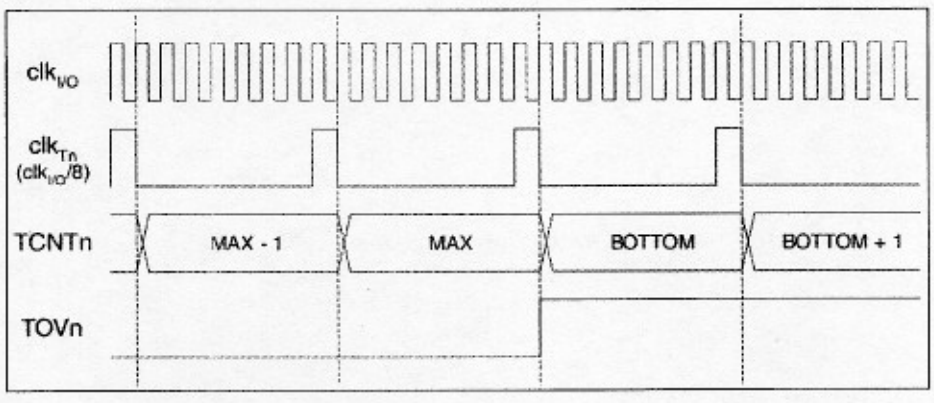
Timer/Counter Timing Diagram, no Prescaling



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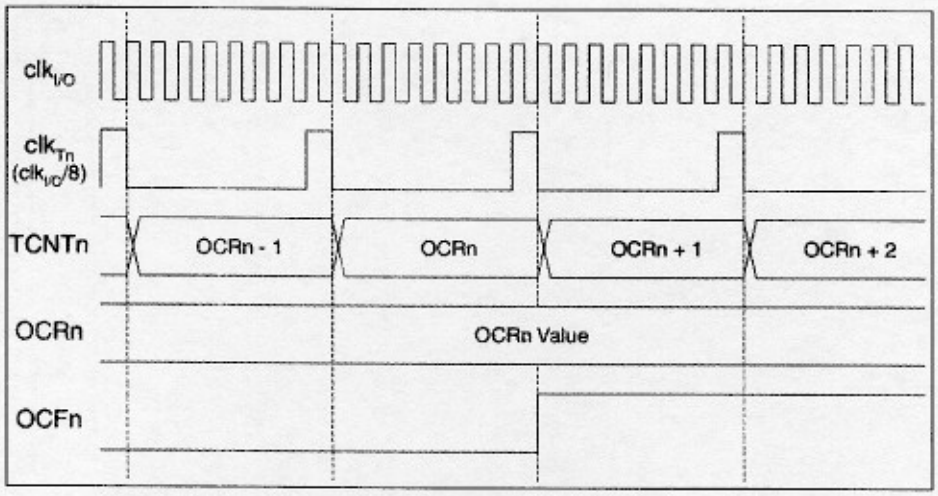
Timer/Counter Timing Diagram, with Prescaler ($f_{clk_I/O}/8$)



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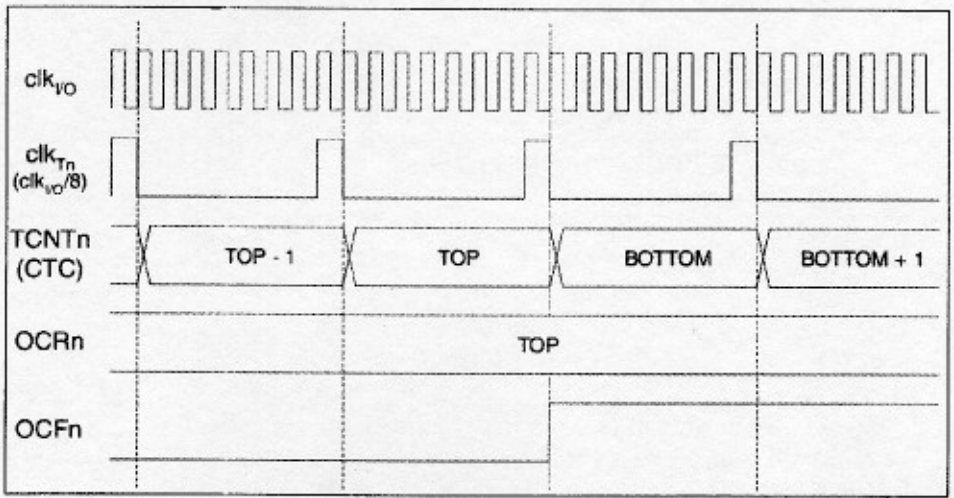
Timer/Counter Timing Diagram, Setting of OCF0, with Prescaler ($f_{clk_I/O}/8$), except CTC mode



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Timer/Counter Timing Diagram, Clear Timer on Compare Match Mode, with Prescaler ($f_{clk_I/O}/8$)



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TCCR0 – Timer/Counter Control Register

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

- Bit 7 – FOC0: Force Output Compare
- Bit 3, 6 – WGM0[1:0]: Waveform Generation Mode
- Bit 5:4 – COM01:0: Compare Match Output Mode
- Bit 2:0 – CS02:0: Clock Select

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TCCR0 – Timer/Counter Control Register

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Waveform Generation Mode Bit Description

Mode	WGM01 (CTC0)	WGM00 (PWM0)	Timer/Counter Mode of Operation	TOP	Update of OCR0	TOV0 Flag Set-on
0	0	0	Normal	0xFF	Immediate	MAX
1	0	1	PWM, Phase Correct	0xFF	TOP	BOTTOM
2	1	0	CTC	OCR0	Immediate	MAX
3	1	1	Fast PWM	0xFF	BOTTOM	MAX

Note: 1. The CTC0 and PWM0 bit definition names are now obsolete. Use the WGM01:0 definitions. However, the functionality and location of these bits are compatible with previous versions of the timer.

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TCCR0 – Timer/Counter Control Register

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Compare Output Mode, non-PWM Mode

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected.
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

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TCCR0 – Timer/Counter Control Register

Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Compare Output Mode, Fast PWM Mode KA_KST,O → Atmel

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected.
0	1	Reserved
1	0	Clear OC0 on compare match, set OC0 at BOTTOM, (non-inverting mode)
1	1	Set OC0 on compare match, clear OC0 at BOTTOM, (inverting mode)

Note:1.A special case occurs when OCR0 equals TOP and COM01 is set. In this case, the compare match is ignored, but the set or clear is done at TOP.

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TIMSK – Timer/Counter Interrupt Mask Register

Bit	7	6	5	4	3	2	1	0
	OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

- Bit 1 – OCIE0: Timer/Counter0 Output Compare Match Interrupt Enable
- Bit 0 – TOIE0: Timer/Counter0 Overflow Interrupt Enable

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TIFR Timer/Counter Interrupt Flag Register

Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

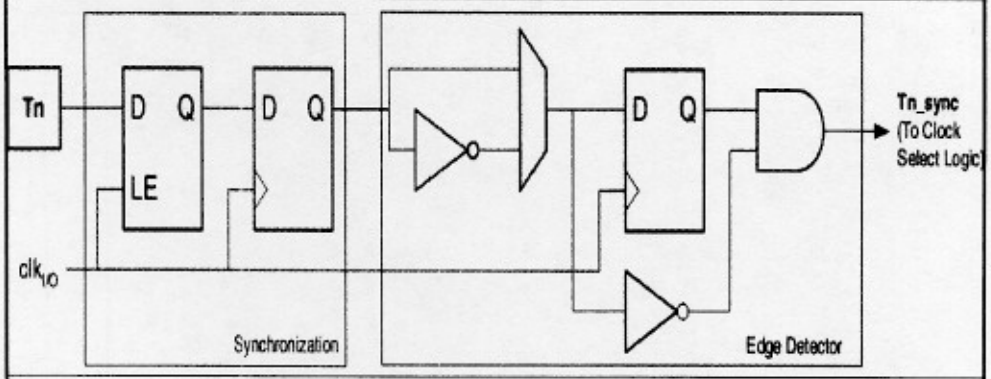
- Bit 1 – OCF0: Output Compare Flag 0
- Bit 0 – TOV0: Timer/Counter0 Overflow Flag

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Timer/Counter0 and Timer/Counter1 Prescalers External Clock Source

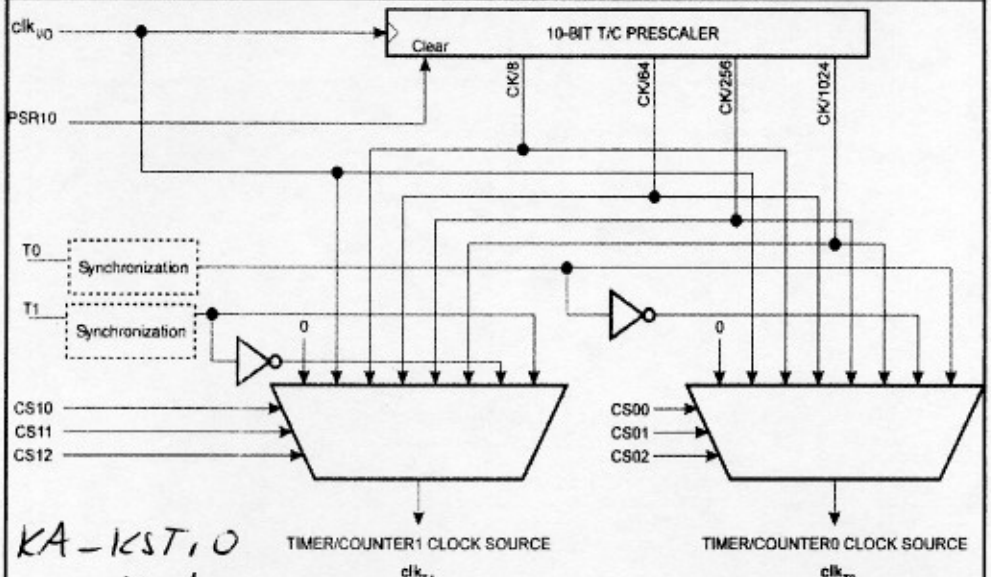
T1/T0 Pin Sampling



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Prescaler for Timer/Counter0 and Timer/Counter1



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